Introduction

For an introduction to the France economy and business cycles, check the appendix.

Updates from the last assignments

Since the last assignment, I have decided to add monthly indicators capturing key dimensions of French economic activity: manufacturing production, manufacturing employment, services employment, and demand evolution in services. These are complemented by three additional cyclical series: started dwellings (reflecting newly approved and initiated residential construction projects), hotel occupancy rates (measuring the share of occupied rooms in the hospitality sector), and air passengers (representing the volume of domestic and international travelers through French airports).

Series	Source	Direction	Sample size
started_dwellings	INSEE	Procyclical	1994M1 – 2026M2
occupancy_rate	INSEE	Procyclical	2003M3 – 2024M2
air_passangers	INSEE	Procyclical	1994M1 – 2025M9
employment_manufacturing	OECD	Procyclical	1962M1 – 2026M2
production_manufacturing	OECD	Procyclical	1962M1 – 2026M2
employment_services	OECD	Procyclical	1962M1 – 2026M2
demand_evol_services	OECD	Procyclical	1962M1 – 2026M2

Since the previous assignment, I have also decided to drop three time series due to data availability:

1. **Construction order books** end in 2016, so it does not match the length of the other time series.
2. **Occupancy rate:** Unfortunately, this time series starts in 2011, which greatly limits the data availability for the coincident indicator.
3. **Car registrations:** Although is a good leading indicator, it starts from 2010 onwards, limiting the data availability for the leading indicator.

Summary of the previous assignments

The chosen models for computing the trend, and the linearization/not linearization decision is briefed below:

variable	model	treatment
export_order_book_level	BSM	Non-linearised
major_purchases_now	BSM	Non-linearised
unemp_exp_12m	LDHR	Linearised
econ_sit_next_12m	LDHR	Linearised
constr_order_books	LDHR	Linearised
construction_conf	LDHR	Non-linearised
car_registrations	LDHR	Non-linearised
services_conf	LDHR	Non-linearised
industrial_conf	LDHR	Non-linearised
order_book_level	LDHR	Non-linearised
stocks_finished	LDHR	Non-linearised
hicp_yoy	LDHR	Non-linearised
started_dwellings	LDHR	Non-linearised
ocupancy_rate	LDHR	Non-linearised
air_passangers	LDHR	Non-linearised
employment_manufacturing	LDHR	Non-linearised
production_manufacturing	LDHR	Non-linearised
employment_services	LDHR	Non-linearised
unemployment_over25	LDHR	Non-linearised
demand_evol_services	LDHR	Non-linearised

Following the methodology used on Assignment 3, linearization of the time series was only applied when the linearized variable significantly improves the overall smoothness of the trend. For all the time series, the linearization method chosen was TRAMO. The model has been selected by maximizing the smoothness of the trend, assessed through visual inspection of the first difference of the trend, as well as the variance of the second difference of the trend.[1]

[1] Used only when the visual representation was inconclusive.

The classification of variables as coincident or leading was based on an analysis of their turning points, visual representations of their trends, and foundational economic theory. The turning points analysis of the selected time series is presented below:

Series	Average Lead to recession start	Average Lead to recession end	Behavior
export_order_book_level	0	-6	Coincident
export_order_book_level	-5	-15	Lagging
construction_conf	1	-9	Coincident
services_conf	4	4	Leading
econ_sit_next_12m	12	-11	Leading
car_registrations	24	-9	Leading
major_purchases_now	9	-8	Leading
industrial_conf	3	-5	Coincident
order_book_level	2	-6	Coincident
stocks_finished	0	-7	Coincident
hicp_yoy	1	2	Coincident
constr_order_books	-1	-10	Coincident
demand_exp_3m	-3	-6	Lagging
started_dwellings	13	-2	Leading
ocupancy_rate	2	-5	Coincident
air_passangers	14	-3	Leading
employment_manufacturing	9	-2	Leading
production_manufacturing	-1	-3	Coincident
employment_services	9	-1	Leading
demand_evol_services	7	-3	Leading

Final selection

After evaluating multiple configurations, the final variable selection for each indicator is as follows:

Leading Indicators

- **started_dwellings:** Number of approved and started residential dwellings. A strong predictor that leads the cycle by **13 months**.
- **services_conf:** Current sentiment in the services sector. Crucial for the French economy, it leads both the start and end of a recession by **4 months**.
- **air_passangers**: number of passengers in flights arriving or departing from France. It leads the beginning of a recession by **14 months**.
- **employment_manufacturing**: a survey indicating the confidence of managers in the manufacturing sector about employment. It leads to the beginning of a recession by **9 months.**

Coincident Indicators

- **hicp**: price index for France, and essential economic variable, which lead the recession by **1 month** on average.
- **order_book_level:** Overall industrial order book levels. Today's orders dictate the current production climate (**2-month** lead).
- **construction_conf:** Current sentiment among construction firms. Moves almost simultaneously with the cycle (**1-month** lead).
- **export_order_book_level (High Level):** Anticipates the pull of foreign demand; when at peak levels, it acts coincidentally (**0-month** lead).

Building the CLI and the CCI

In order to build composite indicators, we will use a non-stationary dynamic factor model following Bujosa, Garcia-Ferrer and de Juan (2013). Let y_t N-dimensional vector of observed time series at time t. For a sample of length T, the full dataset can be represented as an $N \times T$ matrix. The measurement equation is given by:

$$y_t = P\ f_t + e_t$$

Where:

- f_t is a vector of length r, made up of the common non-observable factors.
- P is the factor loading matrix, made up of r columns.
- e_t is a sequence of vectors of idiosyncratic noise. They are assumed to be normally distributed with mean 0 and a covariance matrix of σ_e . Where σ_e is assumed to be a full rank diagonal matrix.

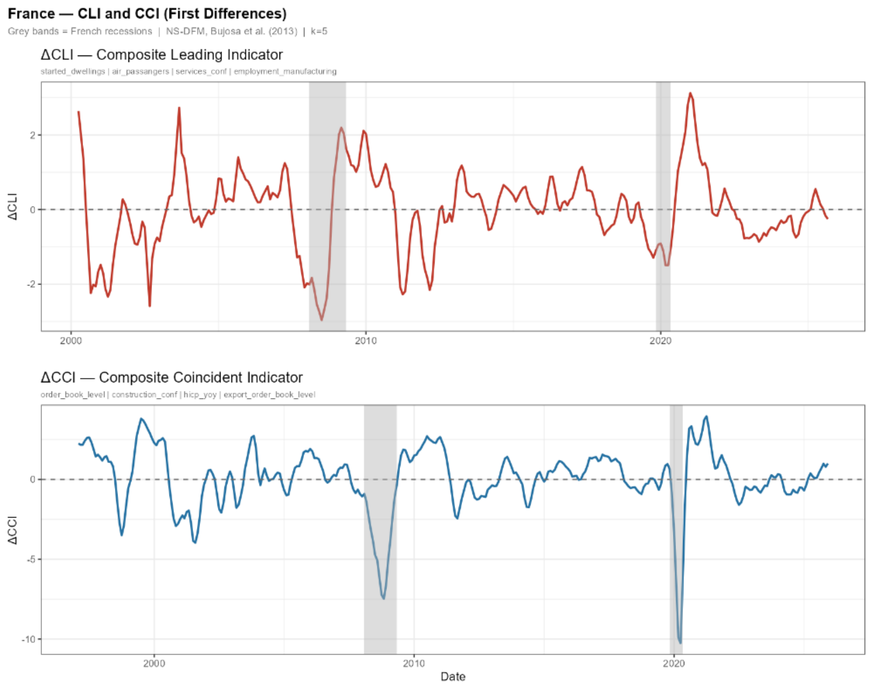
The we assume that y_t is not stationary but of order $I(d)$, where d is the number of regular differences needed to make the time series stationary. The real GDP of France is $I(1)$, meaning that the first difference (and the growth rates) is stationary[2]. Since we have non-stationary factors, the variance covariance matrix would grow to infinity as T increases; to avoid that, we weight the covariance matrix by a scaling factor that neutralises the explosive trend:

$$C_y(k) = \frac{1}{T^{2d+d\prime}} \sum_{t=k+1} (y_{t-k} - \bar{y}\ )(y_{t-k} - \bar{y})'$$

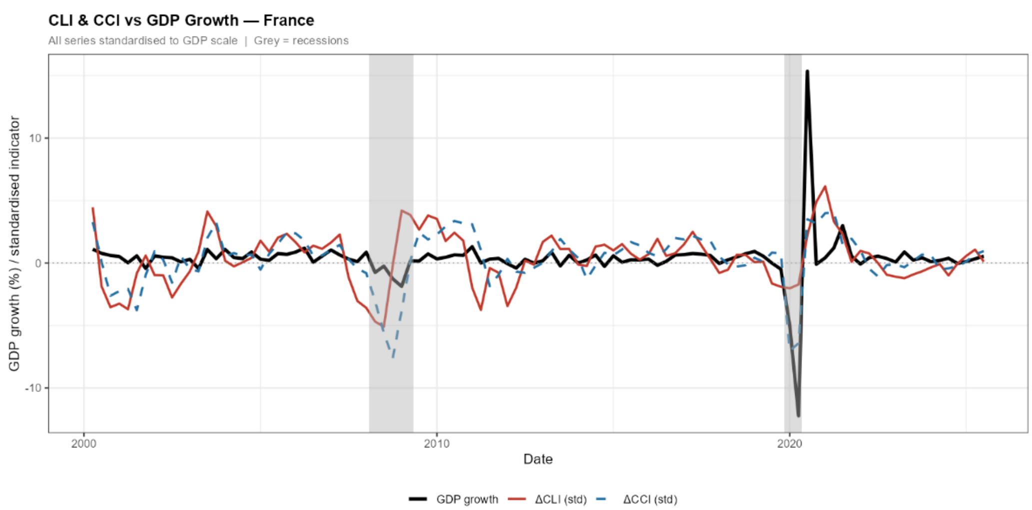
Peña and Poncela (2006) show how $C_y(k)$ converges to a random matrix Γ_y that has at least r pervasive eigenvalues for $k = 5$, so we will follow their specification. The first non-stationary factor (associated with the dominant eigenvalue) becomes our Composite Indicator.

[2] The order of integration was obtained from economic theory and by testing the time series using the Augmented Dickey Fuller test.

The first difference of the resulting composite indicators is the following:



When compared to GDP, we obtain a promising result, especially on the composite indicator.



The composite leading and coincident indicators in levels, and their comparison with GDP are available in Figures 1 and 2 respectively.

The weights of each time series in the composite indicator are presented on the following table:

Group	Variable	Weight
CLI	started_dwellings	0,000991
CLI	air_passangers	0,023466
CLI	services_conf	0,208603
CLI	employment_manufacturing	0,768922
CCI	order_book_level	0,310497
CCI	construction_conf	0,376382
CCI	hicp_yoy	0,004659
CCI	export_order_book_level	0,308462

In order to diagnose our composite indicators, we run 2 types of tests, following Bujosa, García-Ferrer, de Juan and Martín Arroyo (2018) : **factor extraction diagnostics,** which tests if it makes mathematical sense to combine these variables into a single index; and **anticipation diagnostics,** which test if the CLI acts as early warning system of the CCI.

Anticipation diagnostics

The first test is **Barlett's sphericity test,** which tests if a group of variables are entirely uncorrelated.

- The null hypothesis is that all the variables in the correlation matrix of the dataset are the identity matrix (no correlation)
- The alternative is that the correlation matrix is different than the identity matrix.

Although it is a good preliminary test, is highly sensible to the sample size, mainly because the test assumes that the data follows a multivariate normal distribution.

Group	Chi2	Df	p_value	Result
CLI	273,4403	6	0	*** Strong common factor
CCI	1149,9715	6	0	*** Strong common factor

In our case, we decisively reject the null hypothesis for both indicators, confirming that the underlying variables are significantly correlated.

Next, we compute the proportion of variance explained by each common factor (determined by the variance share of its corresponding eigenvalue) to assess how much total variability is captured by our composites.

Group	Eigenvector	Eigenvalue	Share_pct	Cumul_pct
CLI	1	1,567	39,18	39,18
CLI	2	1,5129	37,82	77
CLI	3	0,6721	16,8	93,8
CLI	4	0,248	6,2	100
CCI	1	2,4739	61,85	61,85
CCI	2	0,9748	24,37	86,22
CCI	3	0,523	13,07	99,29
CCI	4	0,0283	0,71	100

Finally, we calculate the Pearson correlation between each individual variable and the extracted composite to ensure no single variable behaves counter to the broader group. (If a variable displayed counter-cyclical behavior, it would require inversion before computing the common factor).

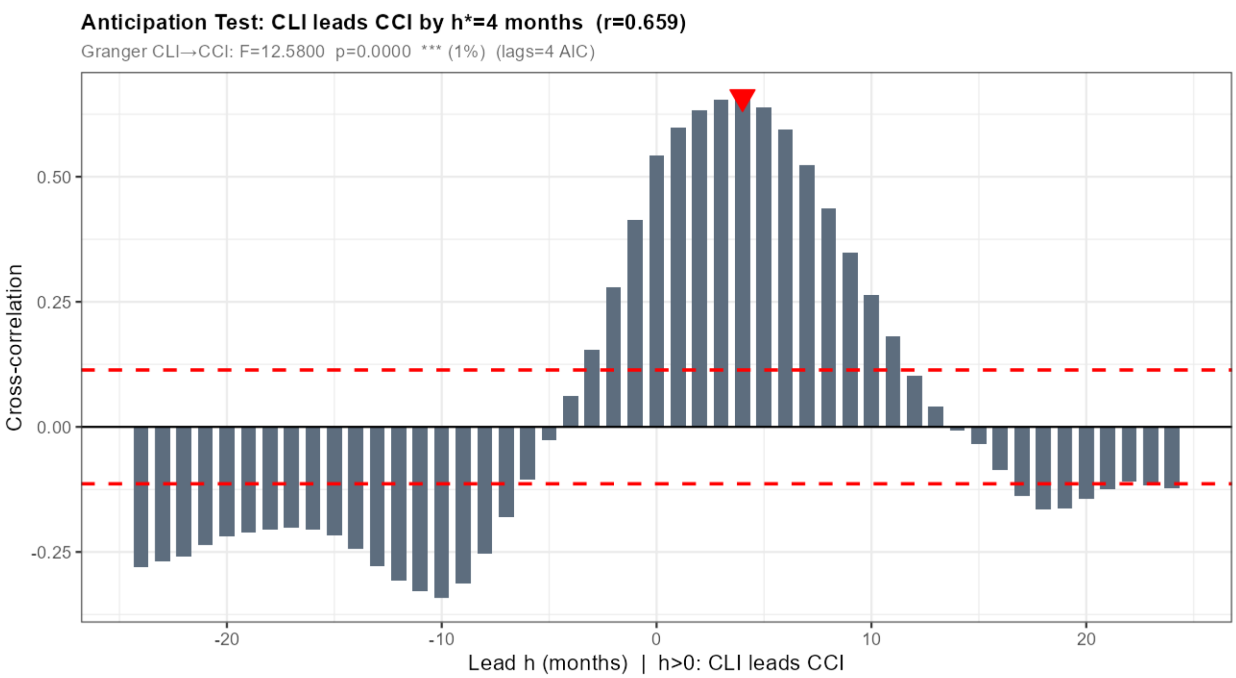
Group	Variable	Corr_with_composite
CLI	started_dwellings	0,0809
CLI	air_passangers	0,2556
CLI	services_conf	0,5688
CLI	employment_manufacturing	0,985
CCI	order_book_level	0,9211
CCI	construction_conf	0,8468
CCI	hicp_yoy	0,3215
CCI	export_order_book_level	0,8493

Anticipation diagnostics

Here, we compare the lags of the CLI against the CCI, computing Cross Correlation Functions and Granger causality tests. The CCF at lag k is computed as:

$$r_{xy}(k) = \frac{\sum(X_t - \bar{X})(Y_{t+k} - \bar{Y})}{\sqrt{(X_t - \bar{X})^2(Y_t - \bar{Y})^2}}$$

The anticipation test states that the highest cross correlation is yielded at lag 4, reaching 0.659. This means that the CLI leads the CCI, anticipating the broader economy with a solid 4-month early warning signal. This limitation is primarily attributable to the unprecedented and unpredictable economic shock caused by the COVID-19 pandemic.



For more detailed cross-correlation functions, check Figure 3,4 and 5.

Moreover, we also run a Granger Causality test, where we estimate 2 equations:

- The first equation puts the CCI as a function of its own lags.
- The second equation puts the CCI as a function of its own lags and the lags of the CLI.

Then we run an F-test on the coefficients of the CLI lags, to see if the lags of the CLI help predict the CCI. We obtain that the CLI granger-cause the CCI, having selected 3 lags (following the AIC information criterion).

Although we don’t have a massive anticipation, our CLI helps predict the economy consistently, so we can say it acts as a 4-month ahead predictor of the economy.

We also run a Granger Causality test of the leading variables over the CCI, obtaining that all the leading variables help predict the CCI at a 1% confidence level. The same test is run but using the CCI variables to predict the CCI, we obtain that only the HICP helps predict the CCI at a 10% confidence level.

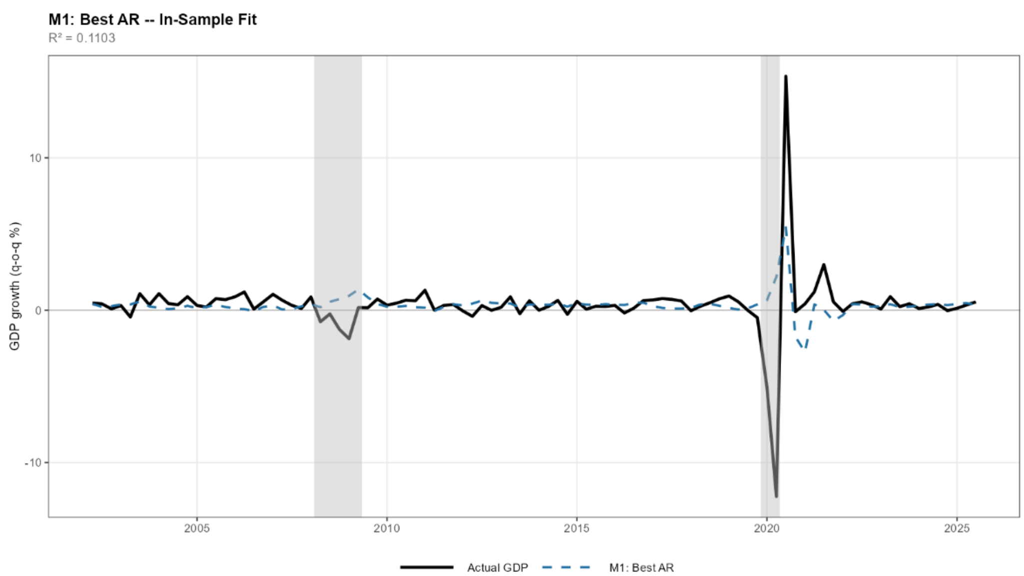
For these reasons, we believe our selection of indicators is highly robust. However, a greater lead time in the Composite Leading Indicator (CLI) compared to the Consumer Confidence Index (CCI) would be preferable.

In-sample specification

Once the composite indicators have been constructed, the analysis proceeds from the monthly factor-extraction stage to the forecasting of quarterly and annual GDP growth. To this end, we adopt a Factor Linear Dynamic Regression model augmented with shock dummies, following the specification proposed by Bujosa, García-Ferrer and de Juan (2013). In line with their approach, we benchmark this model against two simpler alternatives: (i) an AR(p) model, and (ii) a Factor Linear Dynamic Regression without shock dummies.

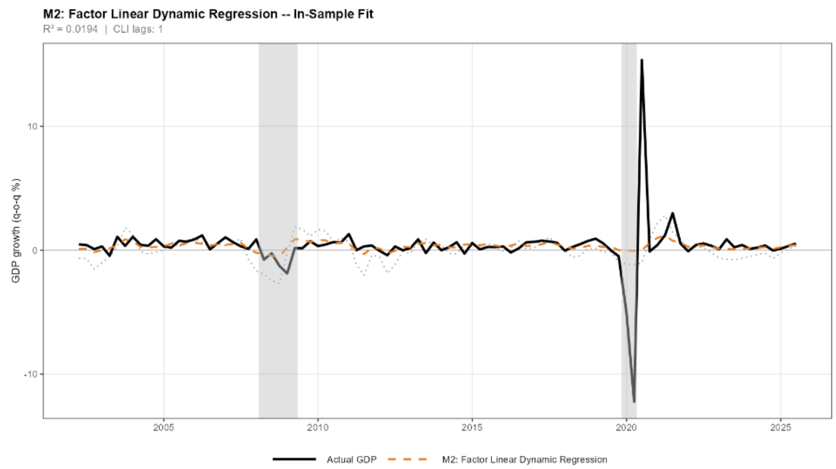
The first model will be an AR model, where we explain the growth rate of GDP as a function of its own lags and a constant. The number of lags to include has been selected by minimizing the AIC criterion (4 lags selected), obtaining an R^2coefficient of 0.11.

$$\Delta GDP = \alpha + \eta_1 \Delta GDP_{t-1} + \eta_2 \Delta GDP_{t-2} + \eta_3 \Delta GDP_{t-3} + \eta_4 \Delta GDP_{t-4} + \epsilon_t$$



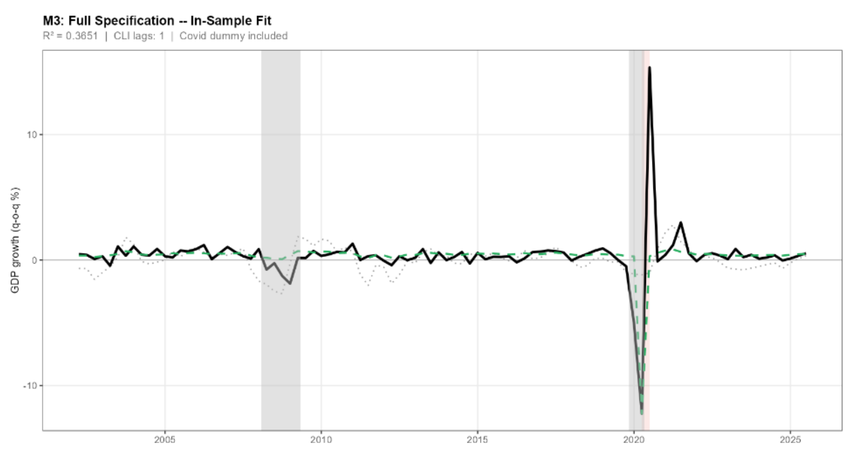
The second model is a Factor Linear Dynamic Regression, where we explain the growth rate of the GDP with the lags of the first difference of the CLI. 1 lag have been selected trying to minimize the AIC criterion, obtaining and $R^2$of 0.02 in the in-sample fit:

$$\Delta GDP = \alpha + \beta_1 \Delta CLI_{t-1} + \epsilon_t$$



Finally, we estimate a full specification model where we include 1 dummy variable for capturing Covid-19 recession. For this equation, 1 lag of the first difference of CLI have been selected by minimizing the AIC information criterion, obtaining an $R^2$of 0.365, and therefore beating the rest of the model in the in-sample fit:

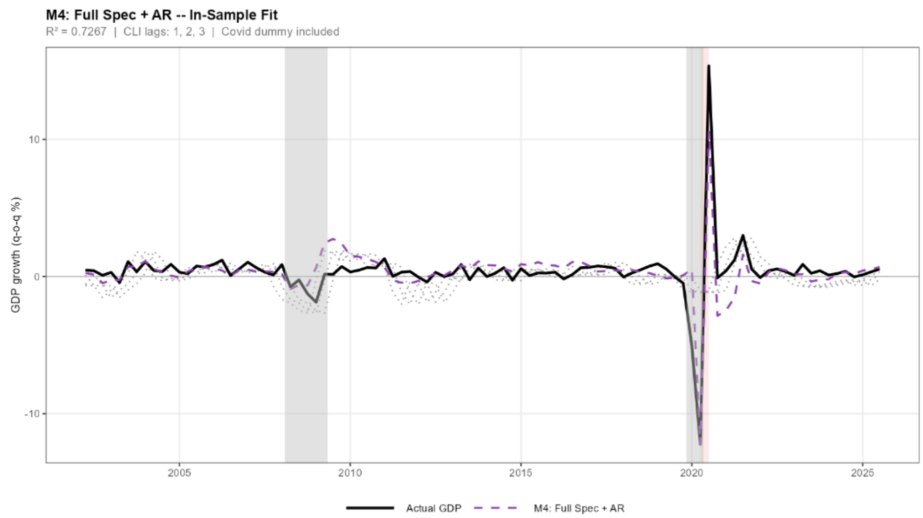
$$\Delta GDP = \alpha + \beta_1 \Delta CLI_{t-1} + \gamma_1 2020M2 + \epsilon_t$$



A better in-sample fit can be obtained by including major_purchases_now in the CLI. In this case, an R^2 of 0.62 would be obtained with 3 lags selected, at the cost of a much worse out-of-sample forecasting.

A 4th model is proposed, where we include a lag of GDP to the third model. With 3 lags of the growth rate of CLI selected, we obtained an R^2 of 0.72.

$$\Delta GDP = \alpha + \eta_1 \Delta GDP_{t-1} + \beta_1 \Delta CLI_{t-1} + \beta_2 \Delta CLI_{t-2} + \beta_3 \Delta \mathrm{CLI}_{t-3} + \epsilon_t$$



Out-of-sample forecasting

The better in-sample fit of the last models (M4,M3) against the simpler models (M1,M2) may be caused due to overfitting. For that reason, it is important to perform an out-of-sample forecasting with the 3 different models to validate the previous results.

The out-of-sample forecasting works as follows:

1. Estimate the models inside the training sample (2000M1 – 2020M12)
2. Do a 1 step ahead forecast (2021M1)
3. Recompute the CLI using 1 more observation (2000M1-2021M1) and recompute the 1 step ahead (2021M2)
4. You do this iteratively until obtaining enough forecasts to compute the RMSE and MAPE, which will be our evaluating metrics.[3]

[3] RMSE and MAPE are jointly employed because they capture different but complementary aspects of forecast accuracy.
RMSE measures the average magnitude of errors in levels and penalizes large deviations, while MAPE evaluates proportional accuracy by expressing errors in percentage terms.
Using both provides a balanced assessment of absolute and relative forecasting performance.

To compare the models forecasts, we need to test if they are significantly different one from each other. For doing so, we employ the Diebold-Mariano test in which we test if the errors of 2 models are significantly different, and therefore, if their forecast is significantly different.

For that, we build a loss function:

$$L_t = (e_{t+h}^1)^2 - (e_{t+h}^2)^2$$

Where $(e_{t+h}^1)^2$ are the residuals from M1 and $(e_{t+h}^2)^2$ are the residuals from M2. Then we run an OLS regression on the loss function using only and intercept:

$$L_{\{t+h\}} = \alpha + u_{\{t+h\}}, \qquad u_{\{t+h\}} \sim N(0, \sigma_u)$$

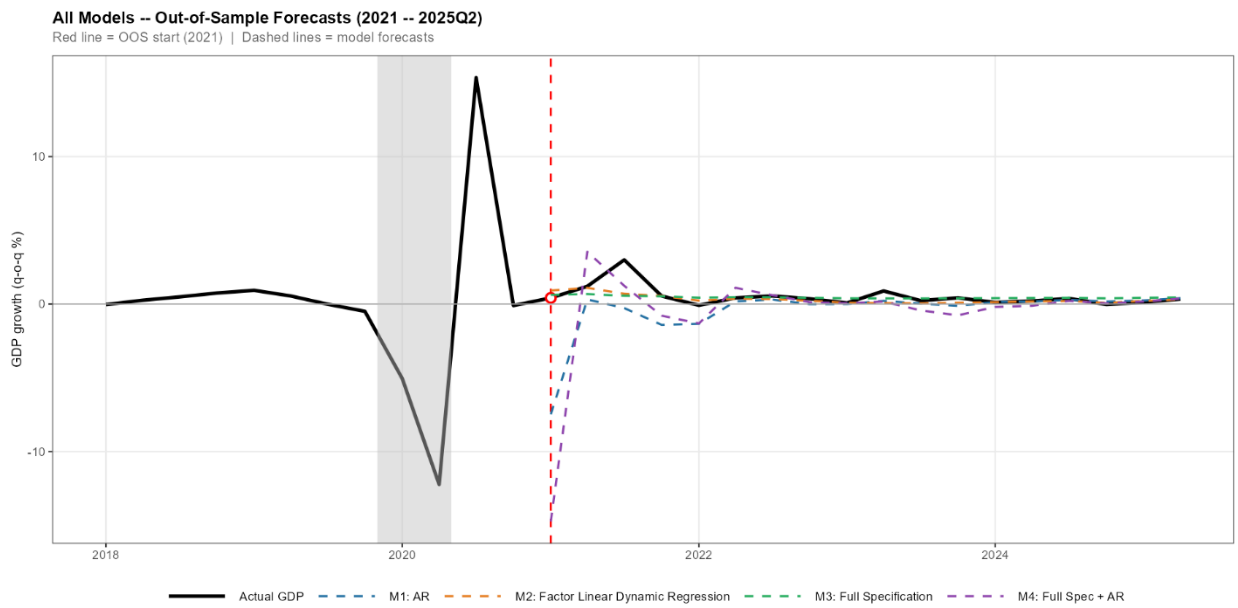
The expected value of the equation is given by the constant α:

$$E[L_{\{t+h\}}] = E[\alpha + u_{\{t+h\}}] = a$$

So if the constant is equal to 0, that means that both forecast errors are virtually the same. For this reason, the test relies on a significant test on the constant of the regression:

- $H_0: \alpha = 0$ (both forecasts are equal)
- $H_1: \alpha \neq 0$ (both forecasts are different)

For the quarterly forecast, the results are presented below:

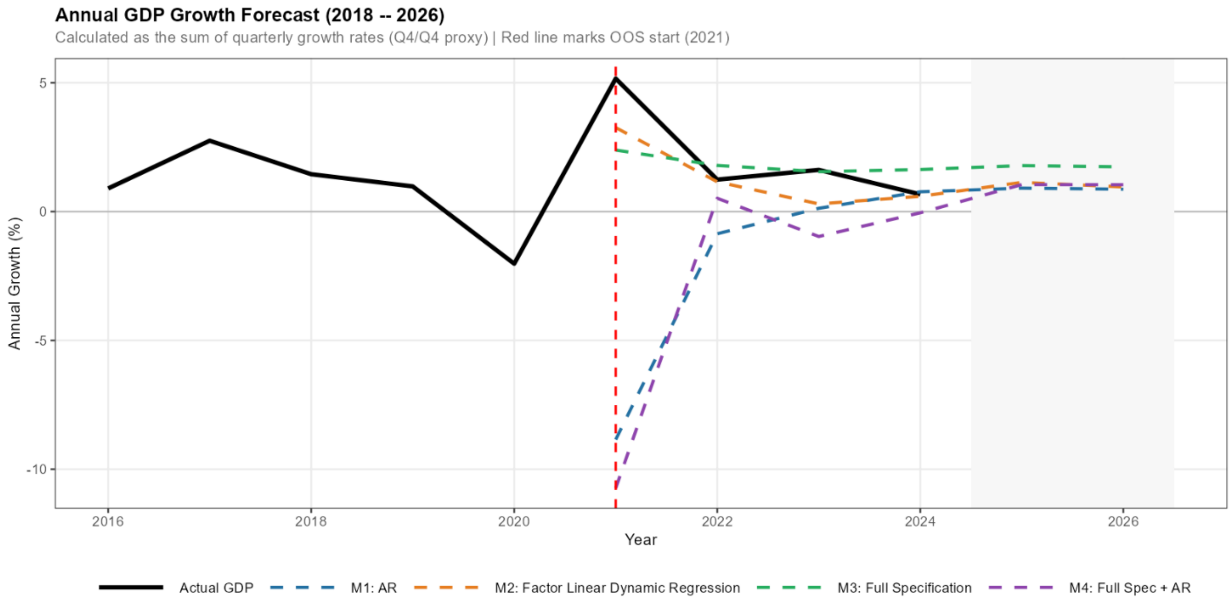


Model	RMSE	MAPE
M1: AR	2,0582	2,8698
M2: Factor Linear Dynamic Regression	0,5902	0,8021
M3: Full Specification	0,6205	1,8537
M4: Full Spec + AR	3,5884	3,8537

The best out-of-sample forecast is archived by the M2 model, but according to the Diebold-Mariano test, the forecasts are not significantly different from each other.

Comparison	Alpha_Intercept	T_Stat	P_Value	Result
M1 AR vs M2 Factor LDR	-0,0191	-0,4738	0,2476	No significant difference
M1 AR vs M3 Full Spec	-0,0111	-1,421	0,2534	No significant difference
M1 AR vs M4 Full Spec+AR	-0,0911	-1,6666	0,3405	No significant difference
M2 Factor LDR vs M3 Full Spec	0,008	0,1944	0,4451	No significant difference
M2 Factor LDR vs M4 Full Spec+AR	-0,0719	-1,6544	0,3123	No significant difference
M3 Full Spec vs M4 Full Spec+AR	-0,08	-1,4044	0,3142	No significant difference

For annual basis, the forecasts are presented below:



In this case, the M2 and M3 models clearly outperform the other models.

The Factor Linear Dynamic Regression (M2) demonstrates strong forecast performance, while the inclusion of a COVID-19 dummy (M3) ensures a robust in-sample fit. By combining this specification with an autoregressive component, the resulting model (M4) achieves better in-sample fit, but the out-of-sample forecast is not significantly improved.

Although the differences in forecast accuracy across these models are not statistically significant, M2 and M3 remain the preferred specification for out-of-sample projections and GDP tracking.

Figures Appendix

Figure 1. Composite Leading Indicator in levels and comparison with GDP (GDP in growth rates and first difference of the CLI standardised)

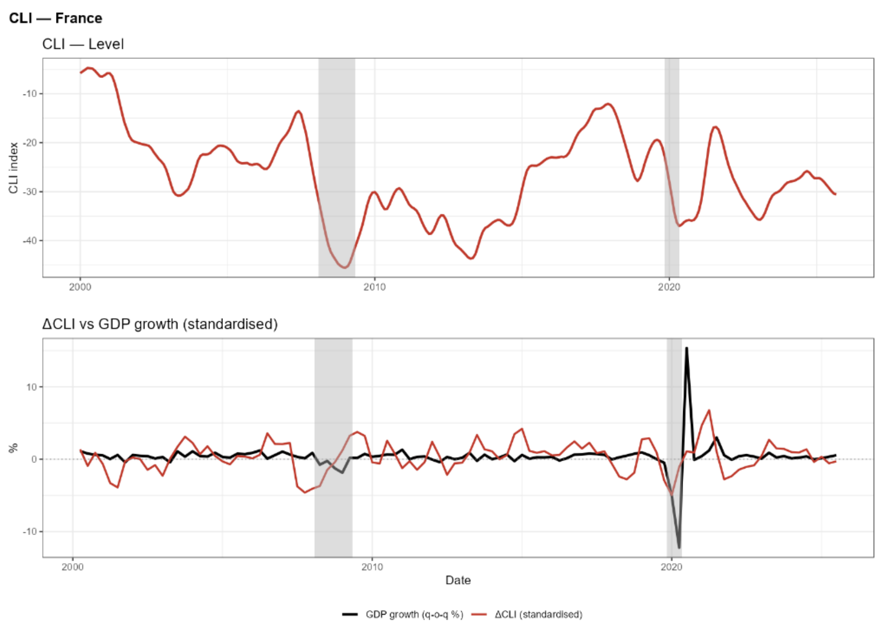


Figure 1. Composite Coincident Indicator in levels and comparison with GDP (GDP in growth rates and first difference of the CLI standardised)

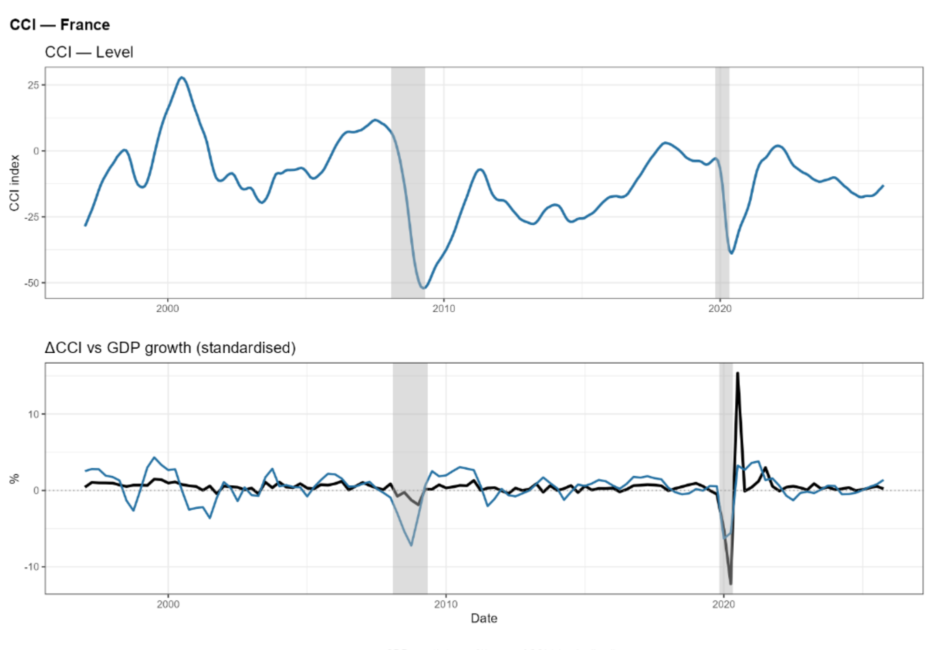


Figure 3. Cross Correlation Function of the leading variables vs the CLI

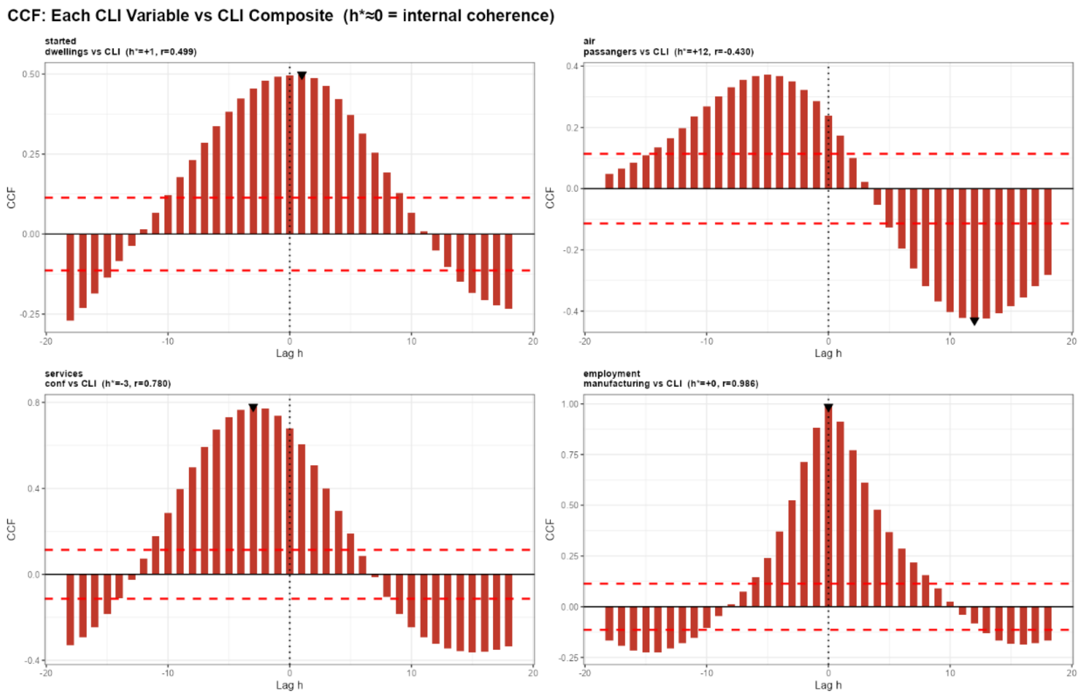


Figure 4. Cross Correlation Function of the coincident variables vs the CCI

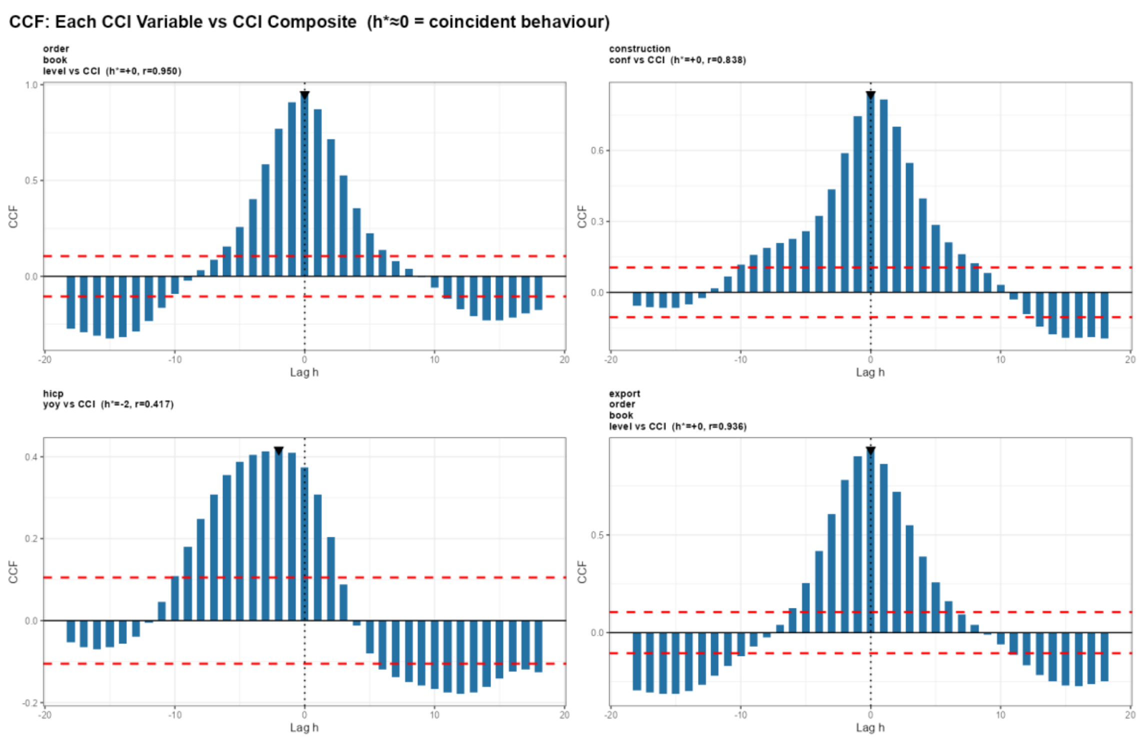
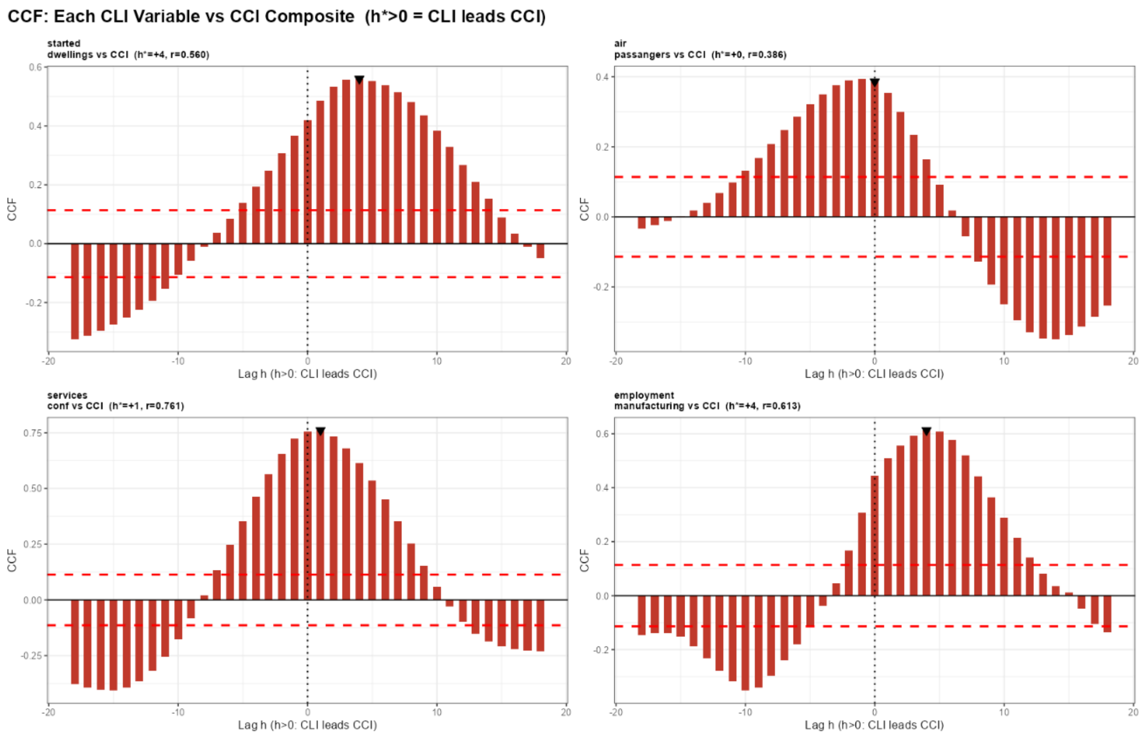


Figure 5. Cross Correlation Function of the leading variables vs. the CCI



Appendix: France Economy

The French economy is the second largest in the Eurozone, in 2026 the country growth was around 1%, driven by household demand and the industrial sector. France exhibits the structure of an advance economy, with the distinctive feature of keeping a leading position across the 3 main activities: Services are the dominant activity (70% of GDP), with tourism and financial services being the main sectors. Industry (18% of GDP) is quite important too, and despite having a smaller share of GDP, it includes some of the most competitive sectors in Europe, like Luxury Goods, Aerospace manufacturing and Automotive manufacturing. Agriculture (2% of GDP) accounts for a small share of GDP, though France is EU's largest agricultural producer, leader in products like wine, dairy and wheat.

As most of EU's large economies, France struggles with a debt overhang, with public debt exceeding 116% of GDP. This problem is specially concerning when measuring the Business Cycles, because high debt loads constraint the use of fiscal policy to counter economic cycles. The unloading of large spending packages (as done during COVID-19) is at risk, potentially making recessions deeper and recoveries slower. A structurally high government deficit (approx. 5,5% of GDP) is added to this problem.

Consumption accounts for over half of French GDP (as will be shown), therefore inflation and wages are critical variables for our analysis. Inflation is targeted by the ECBs monetary policy, so France plays in the same playground as its fellow European countries. We will keep and eye for price shocks in Europe, specially those coming from energy and food prices.

Investment also plays an important role in the GDP creation, so we will consider both he corporate margins, which have been historically low when compared to their European peers, and interest rates, highly related with inflation.

Appendix: France Business Cycles

Created in 2022, the French Business Cycle Dating Committee is responsible for identifying the turning points of the French economy's business cycle and stablishing a historical chronology of the same matter. They define the business cycle attending to the NBER and CEPR definition:

The cycle is defined as a succession of phases of rising activity, i.e., positive economic growth (expansions), and phases of declining activity, i.e., negative growth (recessions). These different periods are delimited by peaks (highest level of activity) and troughs (lowest level of activity), corresponding to the turning points in the cycle.

The methodology used to find this turning point is based on 2 pillars:

1. Quantitative pillar: usage of a series of different econometric tools to produce a list of possibles dates of recession in the French Economy.
2. Qualitative pillar: narrative approach based on the opinions of experts composing the Committee. This pillar aims to validate and filter the quantitative output.

Following this methodology, 5 recession periods are identified. See Figure 1.

1. **1974Q4 - 1975Q2**: due to the 1973 Oil Shock, following OPEC embargo. France was heavily dependant on oil for energy production. This shock had a deeper effect on industries (specially heavy industries), and its effects can be seen on investment, employment and prices (in fact, this was the first stagflation period in France).
2. **1980Q2 - 1981Q1**: following the Iranian Revolution, the world witnessed the second oil shock due to supply disruptions.
3. **1992Q2 – 1993Q3**: European monetary integration trembled when Germany raised interest rates to fight inflation. To keep the exchange rate strong, France had to assume high interest rates. This recession can be observed
4. **2008Q1 – 2009Q3**: the great recession started on the USA, and while French banks were less exposed to "subprime" mortgages than their European counterparts, the recession rapidly spread through Europe, collapsing global trade. As a major exporter, this crisis also reached France borders.
5. **2020Q1 – 2020Q3**: there are debates weather the COVID-19 pandemic is a "forced recession", uniquely because it wasn't cause by the economic environment itself, but by sanitary problems.

References

- **Eurostat:** Provided the Quarterly National Accounts data and the initial monthly confidence and economic indicators.
- **INSEE (National Institute of Statistics and Economic Studies):** Provided specific French monthly series: started dwellings, hotel occupancy rates, and air passengers.
- **OECD (Organisation for Economic Co-operation and Development):** Provided monthly indicators including manufacturing employment, manufacturing production, services employment, and demand evolution.
- **CEPR (Centre for Economic Policy Research) & NBER (National Bureau of Economic Research):** Provided the foundational definitions of business cycles, distinguishing expansions and recessions.
- **French Business Cycle Dating Committee (2022):** Created in 2022; provided the official historical chronology and turning points of the French economy's recessions.
- **Bartlett, M. S. (1950):** *Tests of significance in factor analysis*. Foundational test used to verify if variables in a correlation matrix are entirely uncorrelated during anticipation diagnostics.
- **Bujosa, M., García-Ferrer, A., & Young, P. C. (2007):** *Linear dynamic harmonic regression*. Foundational paper for the LDHR model utilized for time series trend decomposition.
- **Bujosa, M., Garcia-Ferrer, A., & de Juan, A. (2013):** *Non-stationary Dynamic Factor Model*. Foundational paper for the methodology used to build the Composite Leading (CLI) and Coincident (CCI) Indicators, as well as the augmented Factor Linear Dynamic Regression used for GDP forecasting.
- **Bujosa, M., García-Ferrer, A., de Juan, A., & Martín Arroyo, A. (2018):** *Factor Extraction diagnostics and Anticipation diagnostics*. Foundational methodology used for diagnosing the mathematical validity and predictive capacity of the composite indicators.
- **Diebold, F. X., & Mariano, R. S. (1995):** *Comparing Predictive Accuracy*. Foundational paper for the statistical test used during out-of-sample forecasting to determine if the forecast errors between different models are significantly different.

- **Gómez, V., & Maravall, A. (1996):** *Programs TRAMO and SEATS*. Foundational methodology applied for time series linearization, seasonal decomposition, and detection of complex anomalies like the COVID-19 shock.
- **Granger, C. W. J. (1969):** *Investigating Causal Relations by Econometric Models and Cross-spectral Methods*. Foundational paper for the causality test utilized to formally evaluate if CLI lags help predict the CCI.
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